

RAHUL MAJHI

Software Engineer | AI/ML Integration | RAG & LLM Systems | Backend at Scale
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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building AI-powered backend systems, integrating machine learning and NLP into production workflows, and designing scalable data pipelines. Shipped a production AI log analysis system using NLP at Amdocs and am currently designing RepoSage — an AI-powered repository intelligence platform built on RAG (Retrieval-Augmented Generation), vector search, LLM streaming, and AST-based code analysis using Spring AI and Google Gemini. Strong backend foundation in Java, Python, Spring Boot, AWS, PostgreSQL, and distributed systems. Passionate about applying Gen AI to solve real engineering problems.

EXPERIENCE

Software Developer 2 | Amdocs Private Limited

Jan 2026 – Present • Pune, Maharashtra

- ▶ Designed and shipped a production AI/NLP-based log analysis system — applies natural language processing to classify production log entries by severity, surface anomalies, rate issues, and auto-generate GitHub PRs for incident response, significantly reducing MTTR across Kubernetes-based services.
- ▶ Independently built and deployed an Auto Rolldown automation system using algorithmic PR management (conflict resolution, cherry-pick/squash strategies) — adopted org-wide, eliminating 80% of recurring manual effort.
- ▶ Led microservices migration of a large-scale telecom charging platform (Java 17, Spring Boot, Spring Cloud) — designed distributed architecture handling millions of daily transactions with 35% throughput improvement.
- ▶ Architected a Dockerized automation microservice on Kubernetes, integrated into Jenkins CI/CD via REST APIs — demonstrates ability to ship AI-adjacent tooling into production infrastructure end-to-end.

Software Developer | Amdocs Private Limited

Jun 2024 – Dec 2025 • Pune, Maharashtra

- ▶ Applied NLP and ML techniques to build an intelligent log analysis system — first production AI system at Amdocs in this team, establishing patterns for Gen AI integration into backend operations.
- ▶ Provided Level 3 on-call production support for Kubernetes-based systems — maintained observability, performed RCAs, and contributed to operational improvements in a globally distributed environment.
- ▶ Built Jenkins pipeline automation for Linux-based environments and contributed to technical documentation, runbooks, and knowledge bases across the team.

PROJECTS

RepoSage — AI-Powered Repository Intelligence Platform — Personal Project | In Development

- ▶ Designing and building a RAG-based system that ingests any public GitHub repository and enables natural language Q&A, service flow mapping, change impact analysis, and code improvement suggestions — powered by Google Gemini via Spring AI.
- ▶ Implementing hybrid retrieval combining pgvector (vector similarity search) and BM25 keyword search with Reciprocal Rank Fusion — designed to handle Java generics, lambdas, annotations, and nested classes reliably via JavaParser AST chunking.
- ▶ Architecture: Spring Boot 3 backend with SSE streaming for real-time LLM responses, Resilience4j circuit breakers, Caffeine caching, PostgreSQL + pgvector for storage, React + TanStack Query frontend — deployed on AWS + Vercel.
- ▶ Applying advanced design patterns throughout: Strategy (chunking), Visitor (AST traversal), Builder (prompt assembly), Chain of Responsibility (impact analysis pipeline), and Producer-Consumer (@Async ingestion).

AI Log Analysis & Automated Incident Response — Amdocs

- ▶ Built a production NLP pipeline that classifies log severity, rates log quality, and auto-creates PRs for anomaly resolution — integrated into Kubernetes-based CI/CD infrastructure serving enterprise telecom clients.

Serverless REST API — Personal Project

- ▶ Designed and shipped a cloud-native API on AWS Lambda, API Gateway, and DynamoDB with IaC via AWS SAM and CDK — demonstrates end-to-end backend engineering skills supporting AI workload deployment.

Institution Portfolio Platform — NIT Tiruchirappalli

- ▶ Full-stack platform (Node.js, Express.js, React.js) with JWT-based authentication — end-to-end ownership from backend API to frontend, serving real institutional users.

TECHNICAL SKILLS

AI / ML: NLP, RAG (Retrieval-Augmented Generation), LLM integration (Google Gemini via Spring AI), Gen AI tooling

Languages: Java (primary), Python, JavaScript/TypeScript, SQL, Bash, C/C++

Backend & APIs: Spring Boot 3, Spring AI, Spring Cloud, REST API design, Microservices, Apache Kafka, Node.js, SSE streaming

Databases & Vector Stores: PostgreSQL + pgvector, Oracle, MongoDB, DynamoDB — schema design, vector indexing, full-text search

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, S3, IAM, CDK), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD

Resilience & Observability: Resilience4j, Actuator, Micrometer, structured logging, incident management, RCA

Frontend: React.js, TanStack Query, Tailwind CSS, TypeScript

CS Fundamentals: Data structures & algorithms, system design, design patterns, OOP, DBMS, OS, distributed systems

EDUCATION

B.Tech in Instrumentation & Control Engineering | **NIT Tiruchirappalli** *Nov 2020 – Jun 2024 • Tiruchirappalli, India*

ACHIEVEMENTS & RECOGNITION

- ▶ Innovation Appreciation Certificate, Amdocs — awarded for engineering an AI/automation solution that reduced recurring team effort by 80% and was adopted org-wide.
- ▶ Winner, Spider Club Hackathon — NIT Tiruchirappalli; secured 1st place competing against institute-wide teams.
- ▶ Solved 650+ problems on LeetCode and 1,300+ across competitive platforms.